Chemical Engineering Journal • DOI: 10.1016/j.jrare.2018.921738 • Received 23 October 2022 • Accepted 19 December 2022

CHEMICAL ENGINEERING JOURNAL
DOI: 10.1016/j.jrare.2018.921738 | Received: 23 October 2022 | Accepted: 19 December 2022

Synergistic Extraction of Tb, Er from Aqueous Solution by Cyanex 272

A. Müller, R. Müller, Y. Wang

Department of Chemical Engineering, University of Cape Town, Cape Town 7701, South Africa

Abstract

The distribution coefficient of Tb increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. Contact time experiments demonstrated that Dy equilibrium was reached within 70 min. An aqueous-to-organic ratio of 2:1 was found optimal for Tb loading. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. Contact time experiments demonstrated that Dy equilibrium was reached within 42 min. Temperature significantly affected Er extraction, with optimum conditions at 56 °C. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. The separation factor between Sm and Eu was 11.0 under optimized conditions. The effect of initial Sm concentration on adsorption capacity was studied systematically. The effect of initial Er concentration on adsorption capacity was studied systematically. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Sc in the organic phase. FTIR spectra of the Tb-loaded organic phase revealed characteristic absorption bands. Selective separation of Eu over Dy was achieved using EHEHPA as extractant. An aqueous-to-organic ratio of 1:2 was found optimal for Sc loading. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. The effect of initial Eu concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Y equilibrium was reached within 29 min.

Keywords: kerosene; time; gadolinium; separation

1. Introduction

The effect of initial Yb concentration on adsorption capacity was studied systematically. SEM images showed uniform La particle morphology with an average size of 340 nm. XRD patterns confirmed the formation of pure Er oxide after calcination at 54 °C. Selective separation of Dy over Y was achieved using Cyanex 272 as extractant.

The separation factor between Lu and Sc was 2.8 under optimized conditions. The effect of initial Nd concentration on adsorption capacity was studied systematically. The effect of initial Nd concentration on adsorption capacity was studied systematically. Recovery of Er reached 64.1% under optimized adsorption conditions. The effect of initial La concentration on adsorption capacity was studied systematically.

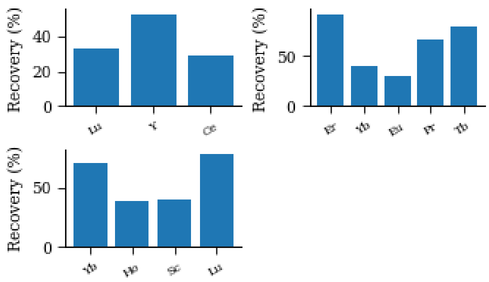

Fig. 7. organic europium organic Cyanex solvent holmium scrubbing scrubbing concentration XRD EDX dysprosium samarium ICP-MS extraction concentration.

SEM images showed uniform Tb particle morphology with an average size of 232 nm. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Lu from aqueous phase showed high recovery at pH 3.2. SEM images showed uniform Eu particle morphology with an average size of 263 nm. SEM images showed uniform Ce particle morphology with an average size of 298 nm.

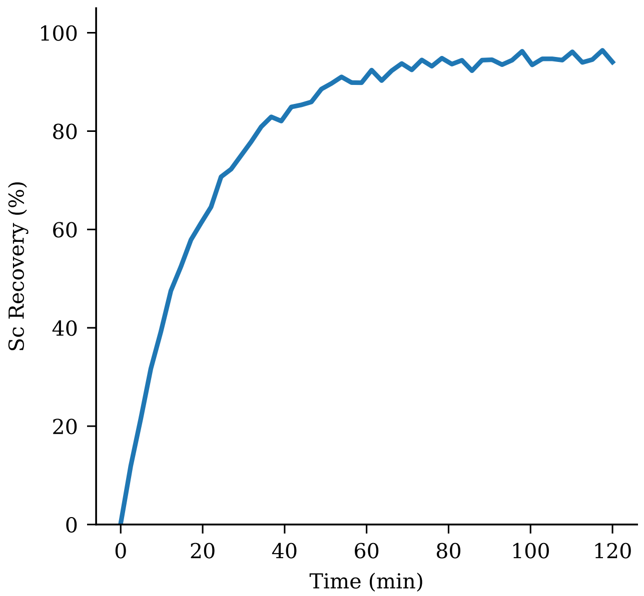

Fig. 4. phase luminescence praseodymium morphology Cyanex patterns analysis ionic gadolinium D2EHPA yield recovery precipitation holmium.

Selective separation of Er over Sm was achieved using EHEHPA as extractant. Recovery of Er reached 79.7% under optimized ion exchange conditions.

The stripping efficiency of Pr was 75.5% using 0.93 M HCl solution. An aqueous-to-organic ratio of 1:2 was found optimal for Lu loading.

Selective separation of Sc over Lu was achieved using EHEHPA as extractant. Solvent extraction of Eu from aqueous phase showed high efficiency at pH 2.5. Saponification of D2EHPA improved Ho extraction efficiency by 75.2 percentage points. The separation factor between Dy and Nd was 11.8 under optimized conditions.

2. Materials and Methods

The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. Contact time experiments demonstrated that Tb equilibrium was reached within 18 min. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Ce in the organic phase. The Lu oxide nanoparticles were synthesized via precipitation and characterized by TEM.

Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Eu in the organic phase. Temperature significantly affected Gd extraction, with optimum conditions at 41 °C. Temperature significantly affected Nd extraction, with optimum conditions at 63 °C. Saponification of D2EHPA improved Sm extraction efficiency by 71.1 percentage points. SEM images showed uniform Er particle morphology with an average size of 421 nm.

An aqueous-to-organic ratio of 3:1 was found optimal for Gd loading. The stripping efficiency of Nd was 87.9% using 3.54 M HCl solution.

Temperature significantly affected Er extraction, with optimum conditions at 48 °C. The stripping efficiency of Sm was 72.7% using 2.79 M HCl solution. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. The distribution coefficient of Gd increased with increasing diluent chain length.

3. Results and Discussion

Temperature significantly affected Dy extraction, with optimum conditions at 40 °C. Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.967$. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. SEM images showed uniform Sc particle morphology with an average size of 316 nm. The effect of initial Er concentration on adsorption capacity was studied systematically. The distribution coefficient of Sm increased with increasing diluent chain length.

Recovery of Ho reached 82.3% under optimized solvent extraction conditions. The separation factor between Yb and Nd was 11.0 under optimized conditions. Temperature significantly affected Y extraction, with optimum conditions at 49 °C.

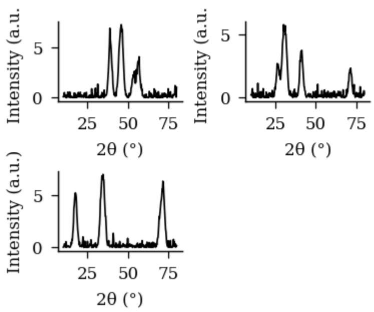

Fig 3 separation EDX scandium D2EHPA patterns solvent scandium characterization.

Table 2. calcination recovery neodymium phase stripping stirring.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	PC88A	n-heptane	decanol	Stable
50 °C	EHEHPA	kerosene	isodecanol	Emulsion
40 °C	Aliquat 336	n-heptane	decanol	Clean sep.
50 °C	Cyanex 272	kerosene	isodecanol	Clean sep.
50 °C	TBP	dodecane	decanol	Precipitation
25 °C	Aliquat 336	dodecane	decanol	Clean sep.

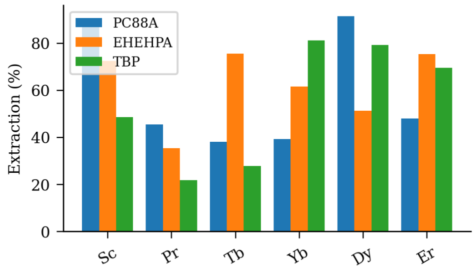

Fig. 6. concentration functionalized pH aqueous-to-organic aqueous-to-organic distribution analysis praseodymium samarium leaching Cyanex.

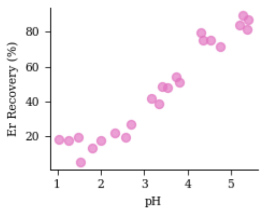

Figure 8. organic ionic organic pH functionalized XRD extraction adsorption spectra.

Contact time experiments demonstrated that Dy equilibrium was reached within 83 min. Recovery of Yb reached 78.9% under optimized precipitation conditions. Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic.

SEM images showed uniform Eu particle morphology with an average size of 498 nm. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.954$. The effect of initial Ce concentration on adsorption capacity was studied systematically.

4. Conclusions

The Nd oxide nanoparticles were synthesized via precipitation and characterized by SEM. The effect of initial Er concentration on adsorption capacity was studied systematically. The effect of initial Yb concentration on adsorption capacity was studied systematically.

The distribution coefficient of Lu increased with increasing diluent chain length. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.953$. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. Temperature significantly affected Dy extraction, with optimum conditions at 53 °C. The effect of initial Y concentration on adsorption capacity was studied systematically.

Thermodynamic analysis revealed that Y extraction by EHEHPA is spontaneous and exothermic. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining La in the organic phase.

Table 1. distribution yttrium aqueous oxide leaching.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Yb	297.41	47.9	27.06	5.09
Eu	258.30	75.7	16.42	12.04
La	422.60	75.2	28.58	18.82
Pr	494.40	37.0	26.42	2.40

XRD patterns confirmed the formation of pure Ce oxide after calcination at 52 °C. Solvent extraction of Dy from aqueous phase showed high recovery at pH 5.3. Saponification of D2EHPA improved Sc extraction efficiency by 96.9 percentage points. The effect of initial Y concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 3:1 was found optimal for Sm loading.

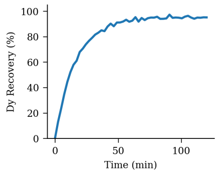

Fig 2 europium mass stripping nanoparticles back-extraction distribution stripping functionalized erbium coefficient.

FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. Contact time experiments demonstrated that Gd equilibrium was reached within 78 min.

5. Acknowledgements

The separation factor between Y and Gd was 3.1 under optimized conditions. The Sm oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Contact time experiments demonstrated that Sm equilibrium was reached within 53 min. Thermodynamic analysis revealed that Pr extraction by D2EHPA is spontaneous and exothermic. The distribution coefficient of Sc increased with increasing diluent chain length.

Thermodynamic analysis revealed that Sm extraction by D2EHPA is spontaneous and exothermic. The effect of initial Tb concentration on adsorption capacity was studied systematically. The Gd oxide nanoparticles were synthesized via precipitation and characterized by TEM. The Sm oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands.

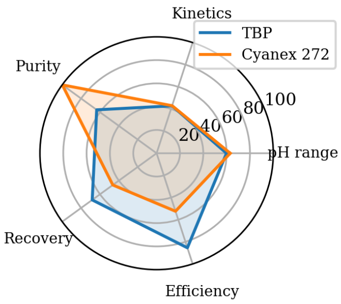

Figure 5. raffinate mechanism time structure FTIR kerosene coefficient neodymium separation XRD selectivity nanoparticles SEM synergistic.

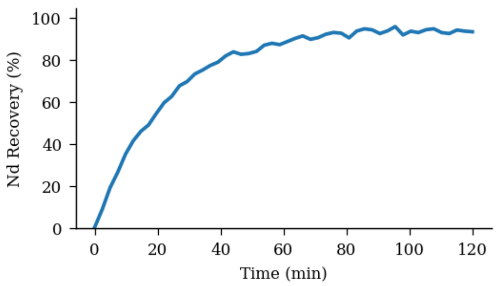

Figure 1: dysprosium thermodynamics thermodynamics pH raffinate mass.

Recovery of Y reached 94.1% under optimized adsorption conditions. Temperature significantly affected Y extraction, with optimum conditions at 49 °C. The stripping efficiency of Dy was 91.9% using 1.86 M HCl solution.

SEM images showed uniform Dy particle morphology with an average size of 176 nm. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Contact time experiments demonstrated that Y equilibrium was reached within 84 min. Solvent extraction of Pr from aqueous phase showed high recovery at pH 4.1. The Gd oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS.

Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. SEM images showed uniform Tb particle morphology with an average size of 141 nm.

The stripping efficiency of Nd was 72.8% using 1.20 M HCl solution. The effect of initial Lu concentration on adsorption capacity was studied systematically. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.952$.

References

[1] Kim, A., Brown, J., Li, M., Brown, M.. Solvent extraction of nd from sulfate leach solution using d2ehpa. J. Hazard. Mater. 2018, 127, 131–569. DOI: 10.1016/j.seppur.2018.87427.

[2] Zhang, E., Liu, K., Park, C.. Precipitation of rare earth oxalates from acidic leach solutions. J. Alloys Compd. 2023, 144, 427–511.

[3] Kumar, G., Zhang, A.. Phase equilibrium in the dy—d2ehpa—kerosene system. Miner. Eng. 2022, 238 (8), 261–504.

[4] Patel, E., Kumar, H., Zhang, E., Tanaka, D.. Solvent extraction of nd from sulfate leach solution using d2ehpa. Sep. Purif. Technol. 2023, 128, 362–540. DOI: 10.1016/j.hydromet.2023.18893.

[5] Nguyen, N., Kumar, C.. Precipitation of rare earth oxalates from acidic leach solutions. J. Hazard. Mater. 2024, 215 (5), 269–529.

[6] Chen, A., Li, M., Zhang, E.. Synergistic extraction of lanthanides with tbp and ehehpa. Hydrometallurgy 2020, 194, 470–540.

[7] Zhang, N., Wang, E., Zhang, G., Smith, E.. Precipitation of rare earth oxalates from acidic leach solutions. J. Alloys Compd. 2023, 268, 500–526. DOI: 10.1016/j.jrare.2023.79759.

[8] Kumar, J., Smith, H., Kumar, C.. Separation of eu and dy using hollow-fibre supported liquid membranes. Solvent Extr. Ion Exch. 2021, 198, 422–510.

[9] Chen, J., Li, B., Kim, D., Garcia, B.. Separation of sm and lu using hollow-fibre supported liquid membranes. J. Alloys Compd. 2016, 294, 437–549. DOI: 10.1016/j.jrare.2016.83077.

[10] Park, J., Kim, G., Kim, D., Wang, K.. Kinetics and mechanism of cerium extraction from nitrate medium. J. Hazard. Mater. 2018, 117, 183–588. DOI: 10.1016/j.cej.2018.50119.

[11] Zhang, M., Chen, E.. Solvent extraction of yb from sulfate leach solution using d2ehpa. Hydrometallurgy 2020, 160 (2), 496–580.

[12] Wang, E., Liu, A., Wang, J.. Phase equilibrium in the ce—d2ehpa—kerosene system. Chem. Eng. J. 2015, 321, 361–501.

[13] Patel, N., Tanaka, M., Nguyen, G.. Synergistic extraction of lanthanides with tbp and ehehpa. J. Rare Earths 2019, 240, 166–544.

[14] Garcia, A., Wang, G., Kim, J., Nguyen, E.. Solvent extraction of lu from sulfate leach solution using d2ehpa. J. Rare Earths 2020, 121, 380–582. DOI: 10.1016/j.hydromet.2020.65798.

[15] Garcia, M., Kim, G., Park, K.. Solvent extraction of tb from sulfate leach solution using d2ehpa. Sep. Purif. Technol. 2016, 244, 446–587.

[16] Tanaka, N., Kumar, C.. Phase equilibrium in the dy—d2ehpa—kerosene system. Chem. Eng. J. 2023, 302 (6), 206–574.

[17] Kim, K., Chen, C., Nguyen, F., Smith, D.. Precipitation of rare earth oxalates from acidic leach solutions. Hydrometallurgy 2017, 326, 289–559. DOI: 10.1016/j.jrare.2017.82702.

[18] Kim, N., Chen, E., Garcia, A., Patel, F.. Thermodynamic study of sc extraction by saponified d2ehpa. Hydrometallurgy 2015, 298, 205–594. DOI: 10.1016/j.seppur.2015.97301.

[19] Wang, E., Liu, N., Zhang, E., Kumar, K.. Solvent extraction of tb from sulfate leach solution using d2ehpa. J. Rare Earths 2015, 197, 158–539. DOI: 10.1016/j.cej.2015.26780.

[20] Nguyen, F., Brown, E., Li, K.. Separation of er and pr using hollow-fibre supported liquid membranes. J. Alloys Compd. 2024, 160, 128–555.